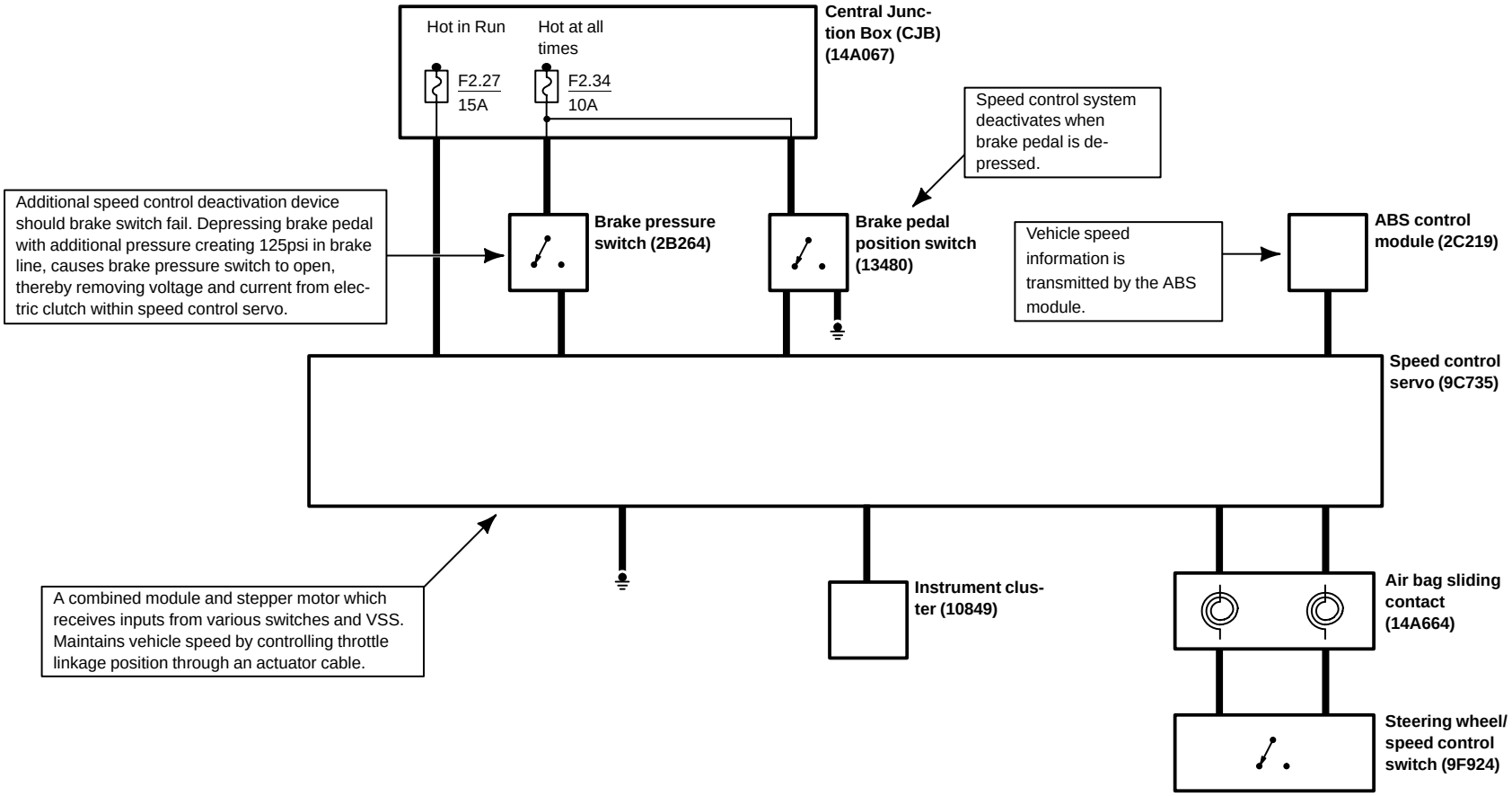System overview, gasoline engines